

SEUNGMIN PARK

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RESEARCH INTERESTS

Large Language Models — retrieval-augmented generation, domain-specific fine-tuning, and agent-based pipelines — and their application to healthcare and biomedical data. Across five first-author papers and industry research spanning clinical LLMs and single-cell transcriptomics, I aim to build language-model systems that turn domain knowledge into reliable, real-world decision support.

EDUCATION

Kyonggi University

Mar 2019 – Feb 2026

B.S., Department of Computer Engineering · GPA 3.9 / 4.5 · Suwon, South Korea

PUBLICATIONS

- [1] Seungmin Park, Angela Galdamez, Dohyup Kim, Yongjin Yoo, PathoBrainSeq R&D Team, & Goonho Park. (2026). Single-Cell Transcriptomic Dissection of Peripheral Immunity Reveals Stage-Specific Signatures in Alzheimer’s Disease: The NeuroMap Platform. Poster Presentation, ADPD 2026.
- [2] Seungmin Park, Minji Kang, Yuchan Kim, Haeyeon Kim, Seokyoung Choi, Seonguk Kim, & Hanyong Lee. (2025). A Study of Dynamic K-Selection and Autoencoder-Based ESG Improvement Activity Recommendation System. Korean Institute of Information Technology (KIIT) Spring Conference. — Gold Award, Undergraduate Research Paper Competition.
- [3] Seungmin Park, Ryeonggu Kwon, Gihwon Kwon, & Changyul Lee. (2025). RAG-Based LLMs for Performance Evaluation and Knowledge Transfer in Software Projects. Korea Conference on Software Engineering (KCSE). — Best Paper Award, Open Competition (all researcher categories).
- [4] Seungmin Park, Suhee Jo, Ryeonggu Kwon, Gihwon Kwon, & Changyul Lee. (2024). LLM-Based Chatbot for Software Project Performance Analysis. Korean Institute of Information Technology (KIIT) Autumn Conference.
- [5] Joonyoung Jo, Seungmin Park, Heesung Kim, Jena Lee, Hyeokju Lee, Catherine Morales, Changjun Kim, & Eric T. Matson. (2024). Evaluating Privacy Infringement Level in Drone-Captured Images Using Privacy Image Quality Assessment (PIQA) Algorithms. Unpublished manuscript.

RESEARCH EXPERIENCE

SELAB, Kyonggi University

Oct 2024 – Feb 2025

Undergraduate Research Intern · Suwon, South Korea

- Built a RAG-based LLM system for a corporate R&D project, vectorizing GitHub development-process data to mitigate knowledge loss caused by high employee turnover.
- Authored two first-author papers from this work; one received the Best Paper Award at KCSE 2025 (open competition, all researcher categories).

Purdue University

Mar 2024 – Jun 2024

Visiting Scholar, IITP Global Education Program · West Lafayette, USA

- Developed a framework to classify privacy-infringement levels in drone-captured images using Privacy Image Quality Assessment (PIQA) algorithms.
- Contributed as a co-first author to the resulting manuscript on privacy assessment for drone imagery.

WORK EXPERIENCE

Gridatech Inc.

Sep 2025 – Jun 2026

Research Engineer · Busan, South Korea

- Fine-tuned a Gemma 4 31B model into a nursing-domain clinical LLM ('Shuri'), building the training corpus from scratch by OCR-digitizing nursing textbooks into paired text/image datasets.
- Built 'ShuShu', a hybrid OCR pipeline for health-checkup forms: benchmarked DeepSeek-OCR, PaddleOCR, and Docling per table layout, routed each form to the best-performing engine, and generated personalized health reports.
- Developed 'Shulingo', an end-to-end real-time speech-to-speech translation service using a RAG-based agent to recognize and translate nursing terminology and abbreviations that general models miss.

PathoBrainSeq

Feb 2025 – Sep 2025

Data Analyst Intern · Palo Alto, USA

- Processed single-cell RNA/DNA sequencing data from Alzheimer's Disease patients in R — applied Harmony integration to correct cross-donor batch effects and produced UMAP/t-SNE visualizations and statistical summaries.
- Delivered analysis-ready outputs to the research team — heatmaps, statistical tables, and summary spreadsheets feeding downstream peripheral immune-signature analysis.
- First-authored a poster at ADPD 2026 introducing the NeuroMap platform.

PROJECT EXPERIENCE

Cost-Efficient Recommendation System for ESG Improvement

Jan 2025 – Jun 2025

Team Leader, Researcher · Suwon, South Korea

- Built a recommendation system for ESG improvement activities — embedded collected ESG data with an autoencoder and retrieved candidate activities via a RAG pipeline with dynamic K-selection.
- First-authored the resulting paper (Gold Award, 2025 KIIT Undergraduate Research Paper Competition) and earned a university capstone award.

Specialized LLM for Personal Statement Editing

Sep 2024 – Nov 2024

Researcher · Suwon, South Korea

- Fine-tuned GPT-4o via OpenAI's fine-tuning API to revise Korean job- and admission-application personal statements, using before/after edits collected from peers as training data.
- Augmented the limited human-edited dataset with GPT-4o-generated examples (Self-Instruct-style) to expand training coverage.

2024 NASA Space Apps Challenge

Sep 2024 – Oct 2024

Team Leader, Researcher · Pangyo, South Korea

- Led a 6-member team building 'SOS: Symphony of Seismics', a deep-learning pipeline to detect seismic events on Earth and Mars from NASA Mars InSight, IRIS, and APSS TWINS data.
- Used a GAN to augment scarce seismic data (13 → 4,000 samples) and a U-Net to denoise signals, factoring in Mars-specific atmospheric conditions. · github.com/2024-NASA-Challenge-Neon-Cactus

HONORS & AWARDS

Best Paper Award, KCSE 2025

Jan 2025

Korea Conference on Software Engineering — open competition, all researcher categories

Gold Award, 2025 Korean Institute of Information Technology (KIIT)

Jun 2025

Undergraduate Research Paper Competition

Gold Award, AWS DeepRacer Campus Competition

Feb 2024

Kyonggi University — reinforcement-learning reward-function design

Merit-based Scholarship, Kyonggi University & Ministry of SMEs ₩10,783,000 <i>Awarded for research excellence and competition achievements</i>	2023 – 2025
Academic Excellence Scholarship, Kyonggi University <i>Awarded for outstanding academic performance</i>	Fall 2023

TEACHING EXPERIENCE

C Programming Mentor — Kyonggi University <i>Mentor · Suwon, South Korea</i>	Mar 2025 – Jun 2025
Teaching Assistant — Basic Capstone Design <i>Undergraduate Student TA, Kyonggi University · Suwon, South Korea</i>	Mar 2023 – Jun 2023

INTERNATIONAL EXPERIENCE

CES 2024 & 2025, Silicon Valley Field Study <i>Team Leader, Student Excursion · Las Vegas & San Francisco, USA</i>	Jan 2024 & 2025
• Attended CES (2024–2025) to track emerging global technology trends; in 2025, led a student team through field visits and networking events with Google, Stanford University, and UKF.	

SKILLS & ADDITIONAL INFORMATION

Programming: Python, R

ML / LLM: PyTorch · LoRA / QLoRA · continued pre-training (CPT) · LLaMA-Factory, Axolotl, Unsloth

Data / Stats: Harmony (batch correction) · UMAP / t-SNE · statistical visualization (heatmaps)

Compute: multi-GPU training on Ubuntu 24.04 — A6000 (48 GB), 2× RTX 5090, A4000 PRO (Blackwell), 4× RTX 3090

Certifications: Engineer Information Processing, SQL Developer, CSTS, Linux Master Level 2

Languages: Korean (Native) · English (New TEPS 391)